

Compositional monitoring of energy-mix drift on the simplex

Which carrier did the structural work?

Not only which carrier got bigger — which carrier moved the composition.

Follow along on the repository — the slide deck, manuscript, and live projector are all open.

H^s runs any compositional dataset the CoDa community can describe — the views in this talk are reproducible on your data.

Operationalizing compositional analysis — a runnable standard for researchers and the AI assistants they choose.

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Community folder: CODA-Association/ · Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR)

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The size view hides the work

A carrier can be small in share and large in structural work.

Germany electricity, 25 years — the standard stacked-area view

Coal and lignite recede. Nuclear phases out.
Gas holds. Solar and wind grow steadily.
Wind passes coal in the late 2010s.

What the size view misses

Germany Solar, 2005 → 2006

starting share	0.21 %
structural Power Share	71.1 %
Activation Coefficient	$\approx 333 \times$

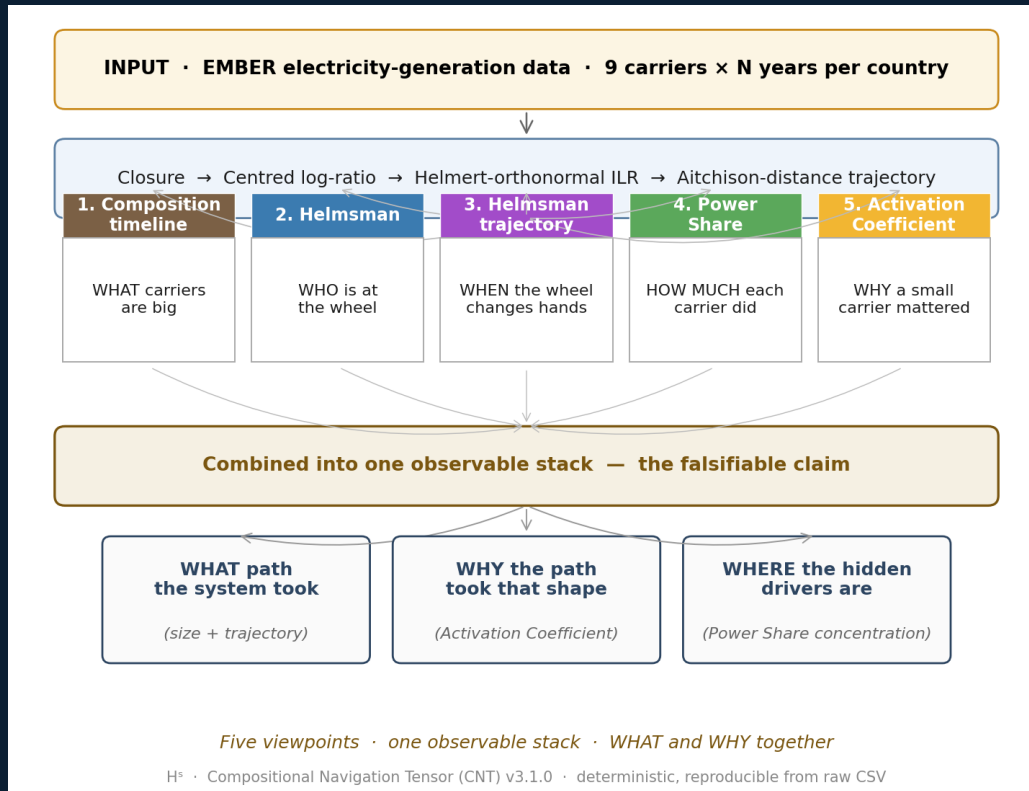
*Solar acted at $333 \times$ its size,
four years before the share view
calls solar visible.*

This talk is the reason that number exists.

Mathematics: standard compositional data analysis. Application: monitoring frame may be new.

Five viewpoints, one observable stack

Each viewpoint answers one question. Together: an auditable transition event.



Composition

what share each carrier has.

Helmsman

which carrier has the largest CLR move at a step.

Helmsman trajectory

when the steering carrier changes.

Power Share

how much squared CLR motion each carrier did.

Activation Coefficient

Power Share ÷ starting share — the yeast factor.

All five derive from CLR + ILR-Helmert. Pure CoDa geometry; no new mathematics.

The Activation Coefficient — the yeast factor

How much structural work a carrier does, relative to how much of the mix it is.

$$\alpha_i(t) = \text{Power Share}_i(t) \div \text{starting share}_i(t)$$

$\alpha \approx 1$ carrier does work proportional to its size — ordinary

$\alpha \gg 1$ carrier acts far above its size — hidden driver

$\alpha < 1$ carrier carries less work than its size suggests — coasting

Worked example — Germany Solar 2005 → 2006

starting share	0.21 %	small
Power Share	71.1 %	most of the work
α	$\approx 333 \times$	yeast

Yeast is 2% of a loaf by mass and does 100% of the rising. Same shape.

The Energiewende's structural beginning, four years before solar appears in the share view.

Three archetypes — one instrument, three regimes

Same protocol applied to three transitions that look fundamentally different.

Germany

deliberate transition

continuous arc

Energiewende
2000 → 2025
solar + wind absorb
structural work
before size dominates.

Japan

external shock

loop and reorganise

Fukushima 2011
displaces nuclear,
causes 2011–2013
multi-year compositional
reorganisation.

United Kingdom

regime change

jump and return

Coal exit 2012–2020
from > 30 % to < 2 %.
Wind, solar, others
absorb displaced
structural work.

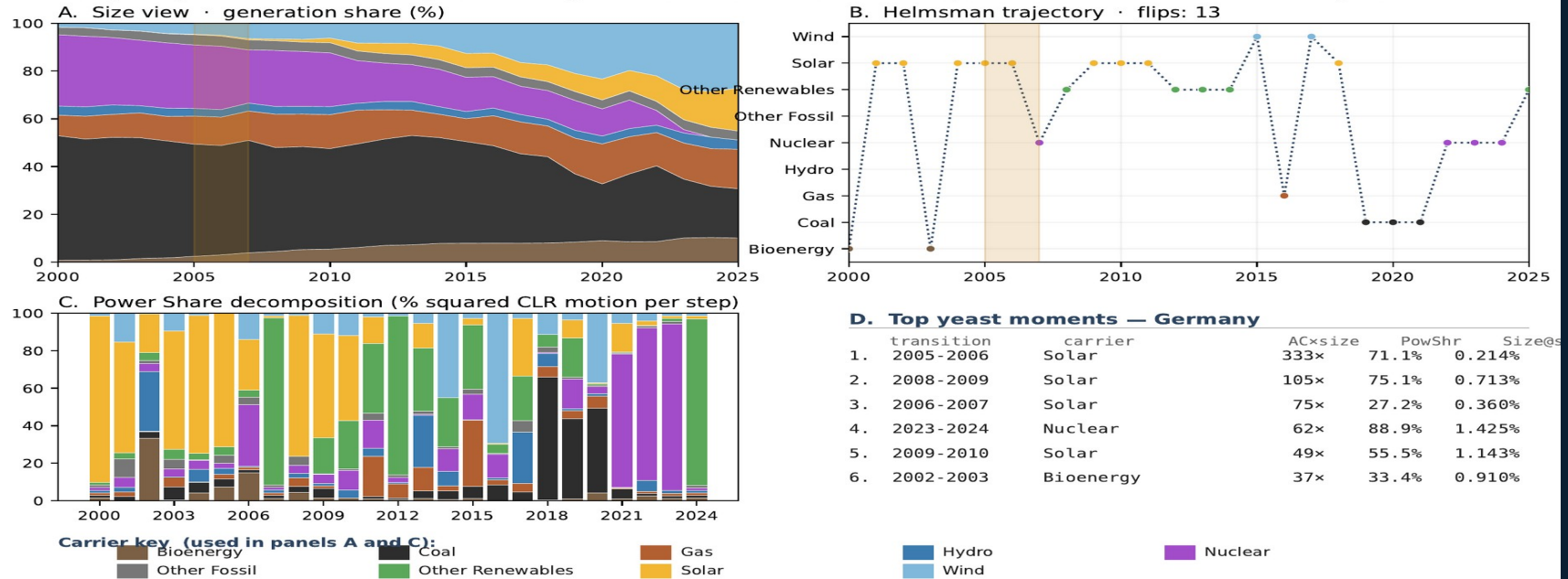
Three different transition regimes. One operational protocol reads them all.

Germany — share-and-work view

Energiewende read as a single smooth arc on the simplex (chart pair: shares + structural work).

Germany — the continuous arc to the renewable vertex

Twenty-five years of deliberate composition change. The size view shift is real — the yeast moments show when it actually accelerated.



Germany's transition is the textbook case of a continuous arc: coal recedes, gas holds, solar and wind grow steadily.

But the Power Share view shows the acceleration moments — solar in 2005-2006 at 0.21 % composition share doing 71 % of the structural directional work (activation coefficient $\approx 333\times$).

That is the structural beginning of the Energiewende — three years before the size view would call it visible.

Hidden Drivers in Energy Mixes · H² CNT v3.1.0 · Aitchison-geometric protocol

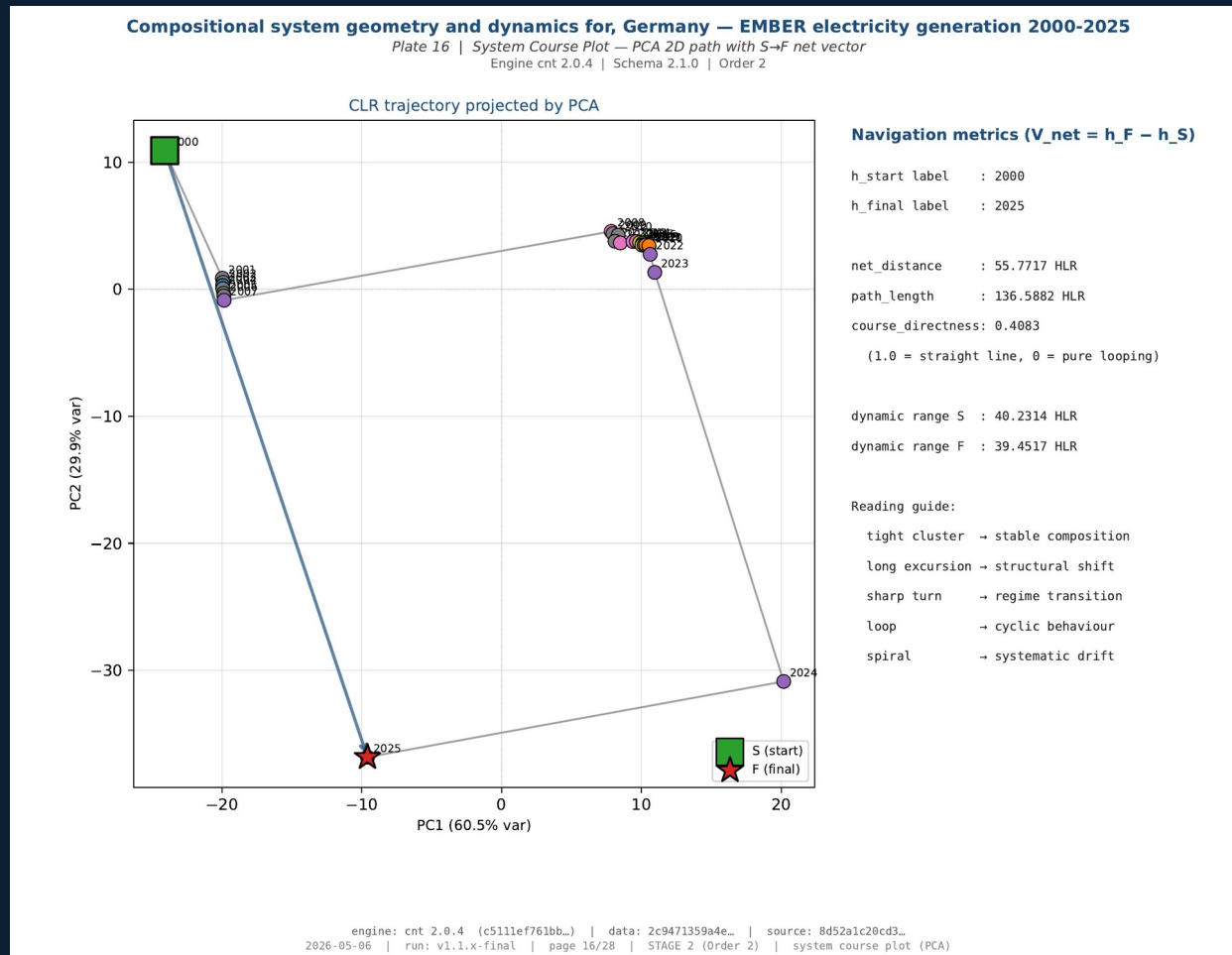
Fig. 2

Solar 2005-2006: 0.21 % share · 71.1 % structural work · $\alpha \approx 333\times$

Tiny share, dominant work. The yeast-factor signature, four years before the share view shows anything.

Germany — course on the simplex

The Helmsman trajectory: where the composition went, year by year.



Course directness 0.41 — continuous arc toward the renewable vertex.

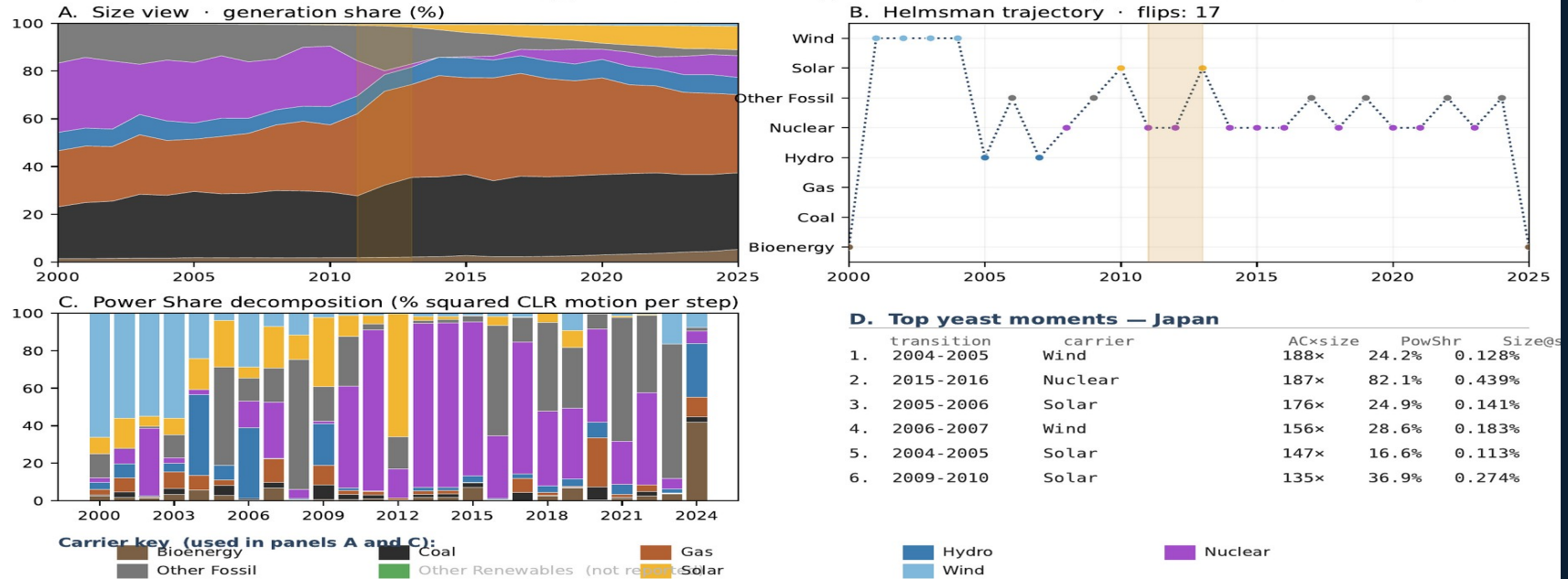
Smooth, monotone reorientation — no loops, no flips. Deliberate transition as a sustained course.

Japan — share-and-work view

Fukushima 2011: external shock displaces nuclear, multi-year compositional reorganisation.

Japan — Fukushima 2011 and the structural aftermath

An external shock and the multi-year recovery path. The helmsman flipped 17 times across the period — the loudest in the corpus.



The 2011-2012 transition shows in every viewpoint at once: composition (nuclear collapses), helmsman (carriers shift abruptly), Power Share (massive concentration of squared CLR motion at one step).

The years AFTER tell the deeper story — Japan's post-Fukushima mix kept reorganising for over a decade, and the yeast moments distribute across multiple carriers as solar, gas, and renewables each took turns absorbing the displaced nuclear share.

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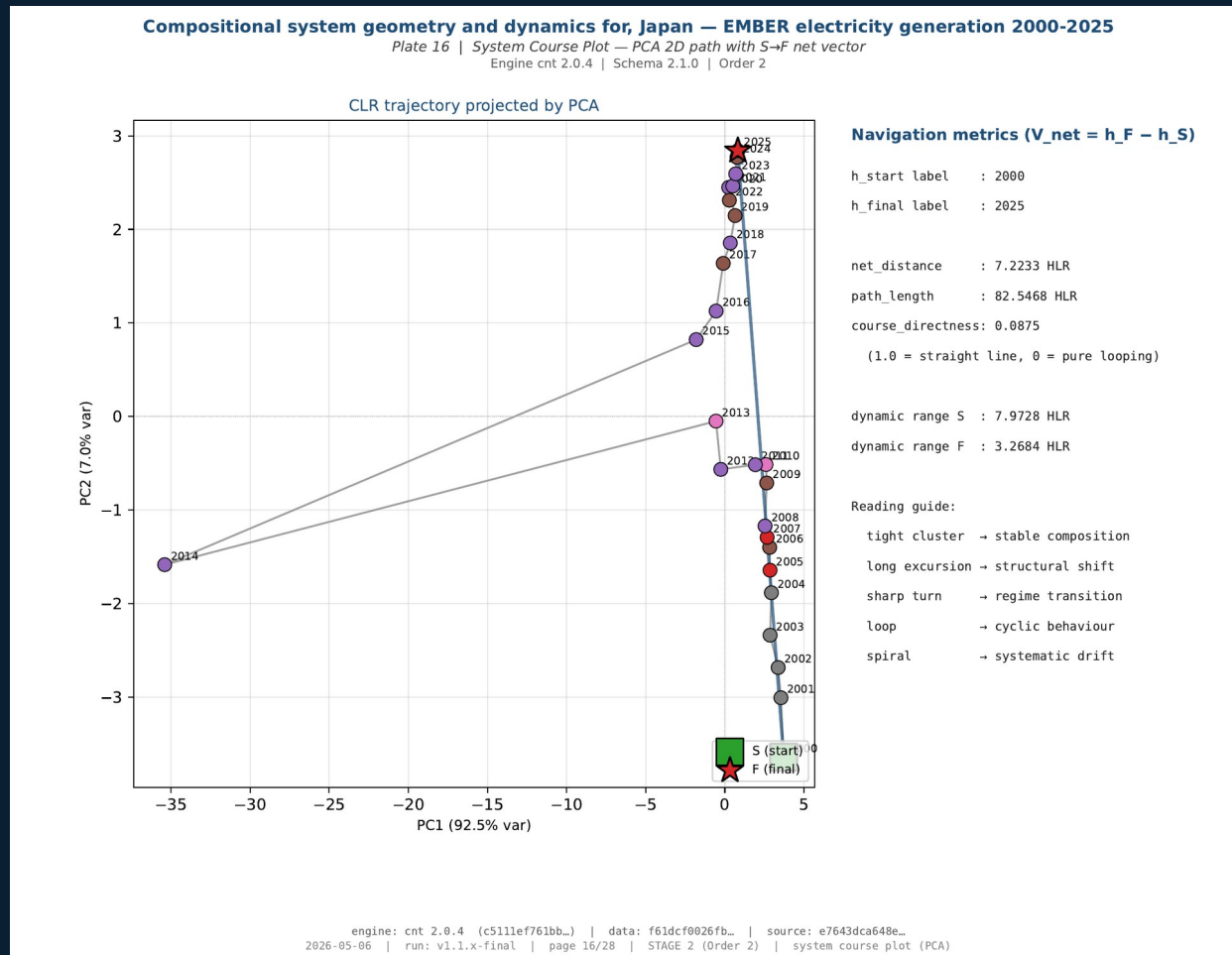
Fig. 3

Aitchison distance 2011 → 2012 ≈ 3 × neighbouring-year baseline · helmsman flips 17 ×

The instrument detects both the shock and the multi-year reorganisation that followed.

Japan — course on the simplex

The Helmsman trajectory: looping reorganisation, not a single step.



Course directness 0.09 — loop-and-reorganise archetype.

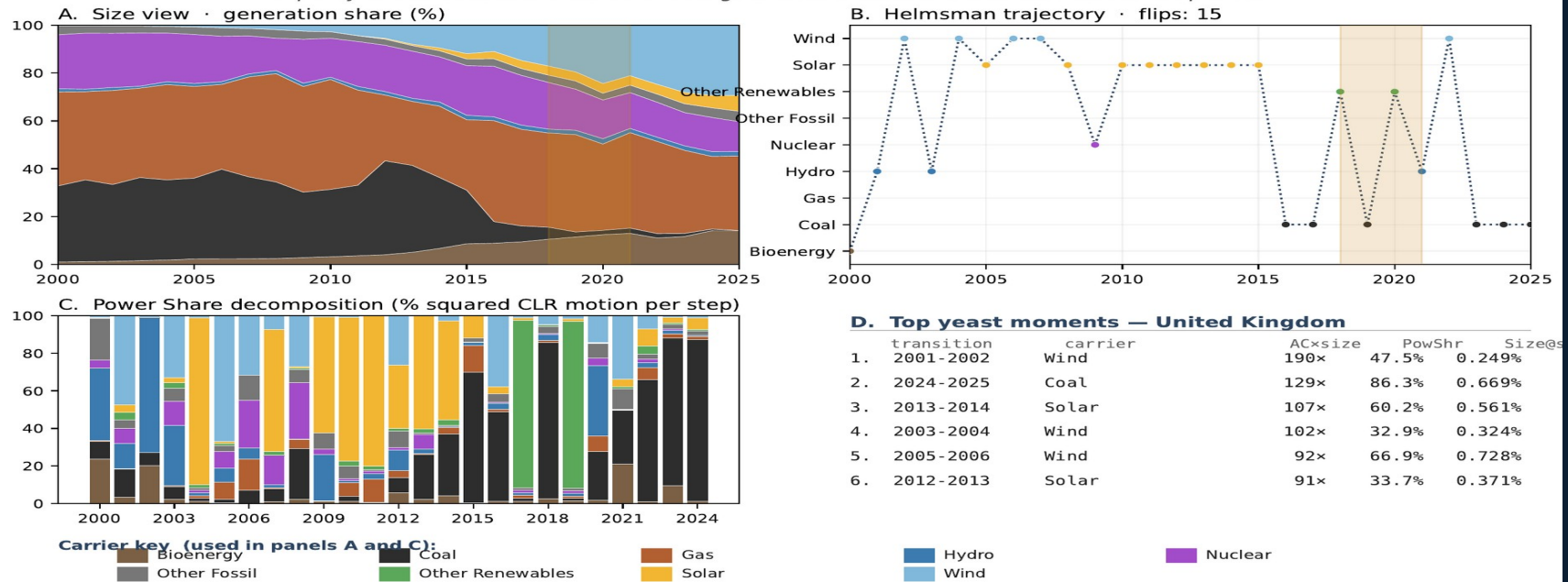
Trajectory revisits and reroutes — the system searches for a new composition, not a planned course.

United Kingdom — share-and-work view

Coal exit as policy-driven regime change, absorbed across multiple renewable carriers.

United Kingdom — the coal exit as a regime change

A policy-driven structural transition. Coal goes from dominant to near-zero across the period.



The UK story is the abrupt-policy archetype. Coal's share drops from over 30 % to under 2 % across the period.

The Power Share view names what made it possible: Other Renewables and Wind in the late-2010s took turns doing the structural work of absorbing coal's exit, often at low single-digit composition share (panel D).

Without the activation coefficient, those small-share carriers are invisible. With it, the whole transition snaps into focus.

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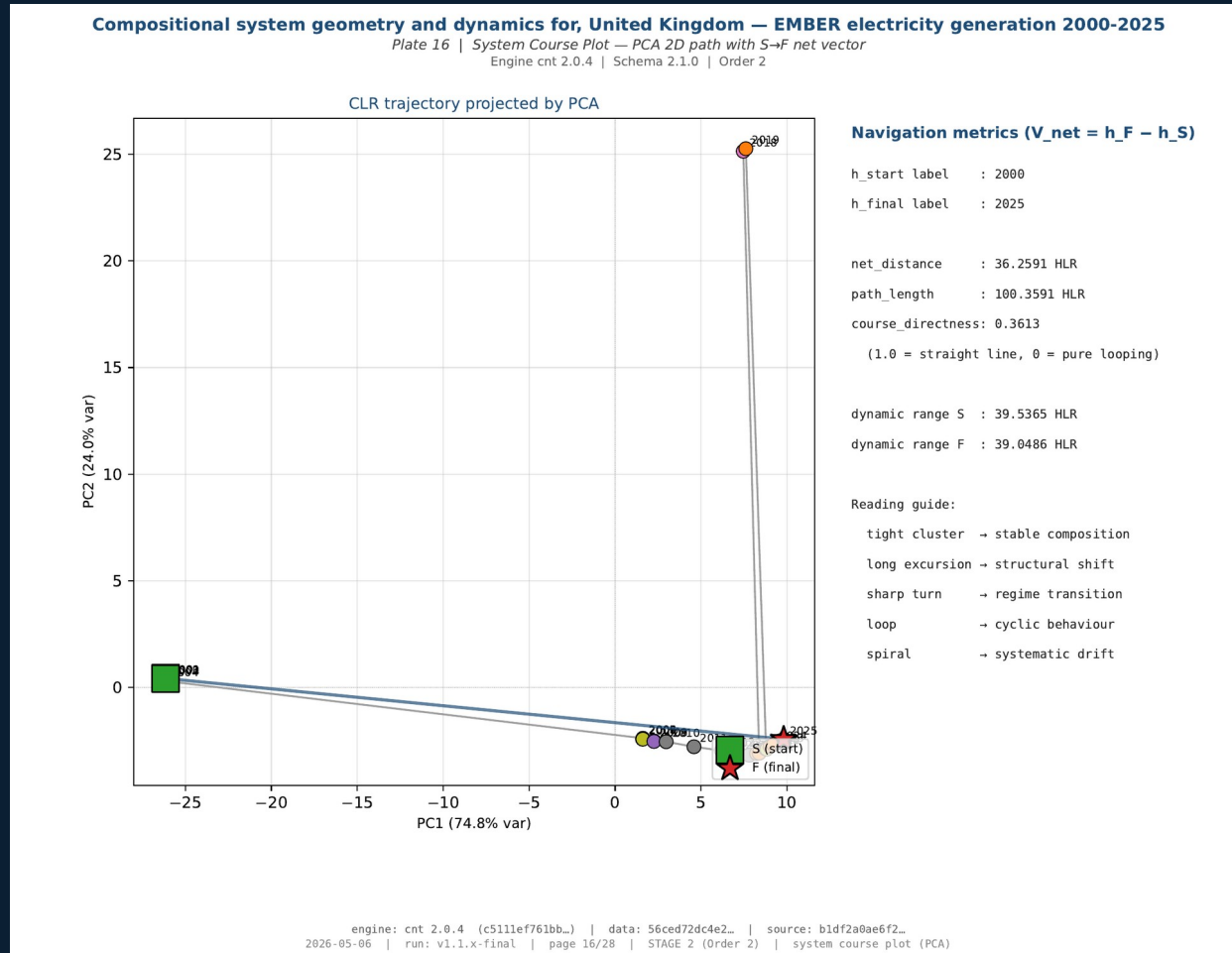
Fig. 4

Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced structural work.

Displaced work redistributes across several carriers — not one substitute, but several.

United Kingdom — course on the simplex

The Helmsman trajectory: a jump from the coal vertex, then a return toward stability.

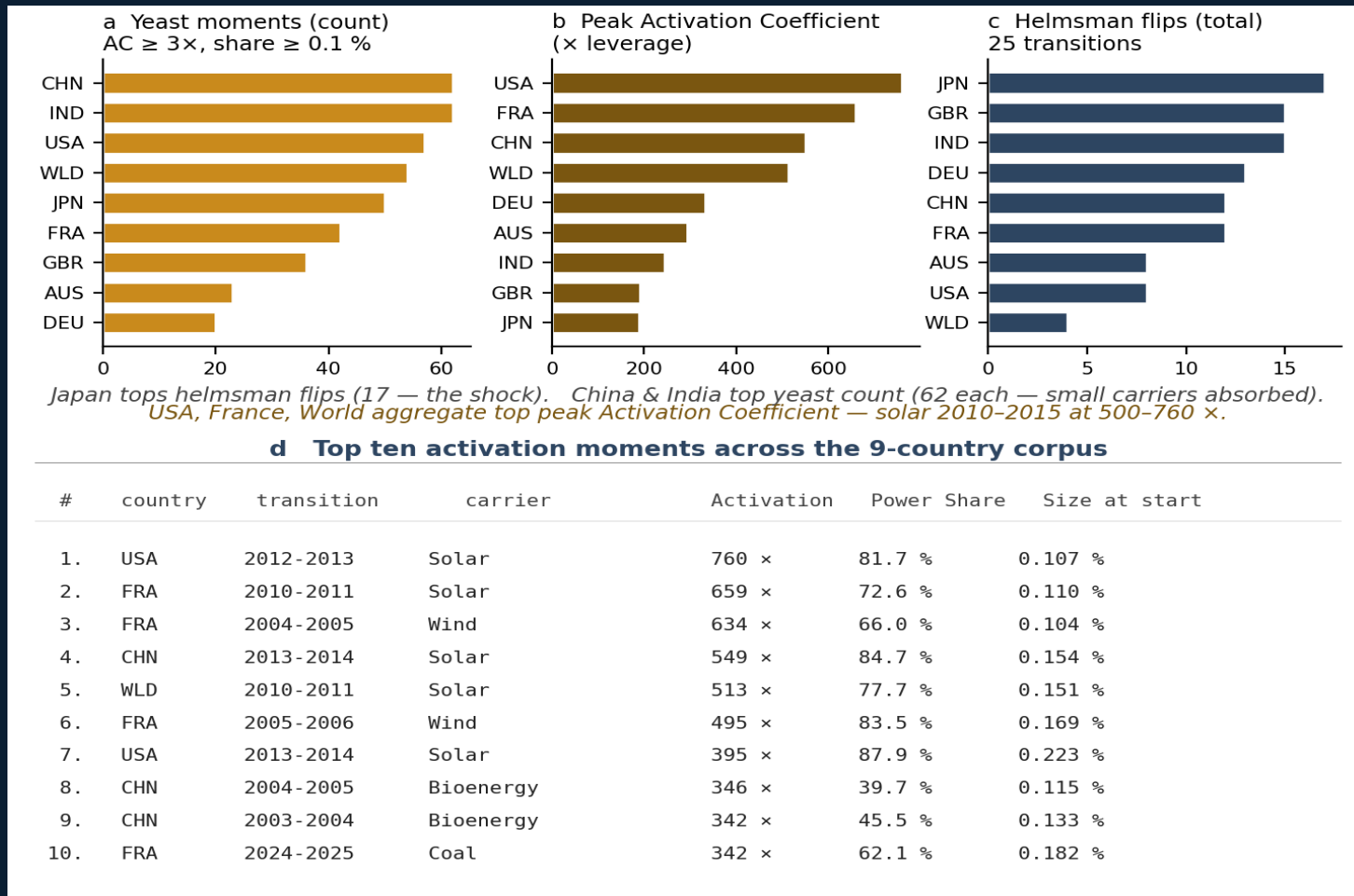


Course directness 0.36 — jump-and-return archetype.

Sharp departure from the coal vertex, then re-stabilisation — regime change as one decisive displacement.

Cross-country signature — 5 of 9 reproduce the pattern

From three case archetypes to a corpus-level result.



fires AUS • CHN • GBR • IND • JPN

quiet DEU (annual) • FRA • USA • WLD

A useful detector should not fire everywhere. Discrimination is itself evidence.

What the stack answers

One observable, five distinct questions, one reproducible object.

WHAT	carriers are big.	<i>size view</i>
WHO	is at the wheel.	<i>Helmsman</i>
WHEN	the steering changes.	<i>Helmsman trajectory</i>
HOW MUCH	work each carrier did.	<i>Power Share</i>
WHY	a small carrier mattered.	<i>Activation Coefficient</i>

The stack does not replace interpretation. It gives interpretation a reproducible object.